



@ieExplained

#ExpressExplained
If there are questions of current or contemporary relevance that you would like explained, please write to explained@indianexpress.com

• NOBEL PRIZE IN PHYSICS

Through ‘ghost particles’ in ice, window to universe



AMITABH SINHA

NEUTRINOS, VERY small sub-atomic particles produced mainly in nuclear reactions in stars, are among the most abundant particles in the universe, second only to photons (or light particles). And they can reveal a huge amount of new information about the universe. The problem has been that they are extremely difficult to detect, the reason they are often described as ghost particles.

The quest to detect neutrinos, and attempts to study them, have already led to several Nobel Prize wins. Another one was added on Tuesday, with the 2026 Nobel Prize in Physics being awarded to Francis Halzen, a Belgian-American scientist who designed an elaborate set-up under ice sheets in Antarctica to detect these elusive particles.

The IceCube facility in Antarctica is not the first observatory to detect neutrinos. Neutrinos were first proposed in the 1930s and first detected in the 1950s. Several observatories around the world are dedicated to detecting and studying these particles.

Studying distant space

What the IceCube facility managed to do was something unique. It was successful in detecting a special variety of high-energy neutrinos that was coming from far away in space. Till now, only those neutrinos that originated in the close vicinity of the Earth, primarily being produced in the Sun, were being detected. Most of the neutrinos passing through the Earth are produced in the nuclear processes inside the Sun.

For scientists, the detection of extra-galactic neutrinos has opened a new source of information about the universe. Initially, visible light was the only tool available to scientists to study the cosmic phenomena. Then, detectors were developed for the full electromagnetic spectrum of radiation and for cosmic rays. The detection of gravitational waves in 2015 opened an entirely

• ‘For decisive contributions to the IceCube Neutrino Observatory and the discovery of high-energy neutrinos of astrophysical origin’

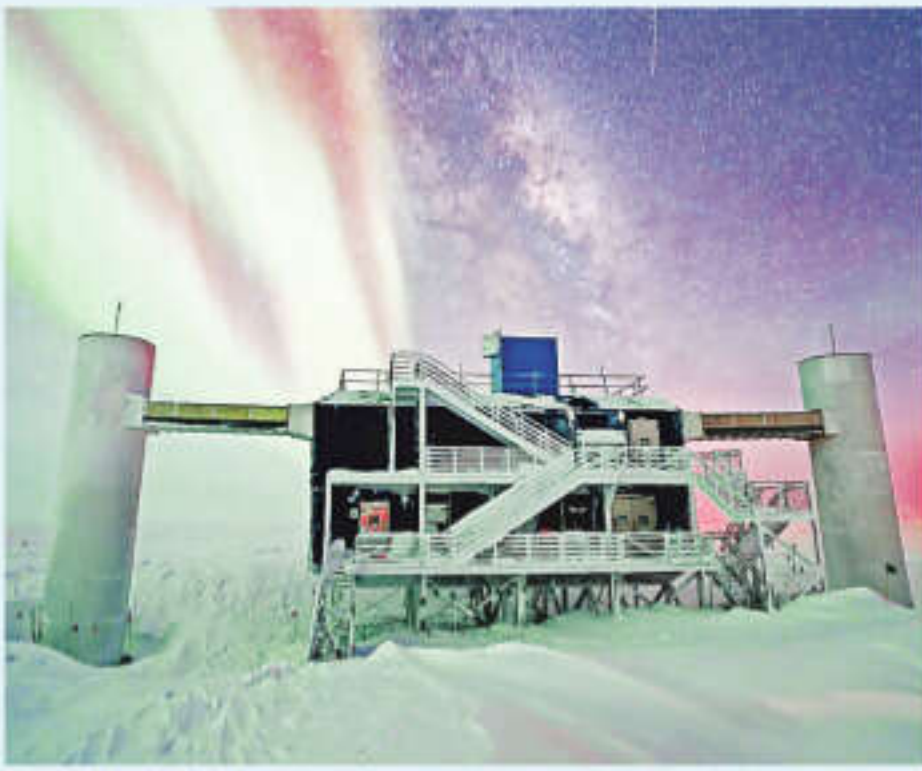


“His idea was to use glacial ice at the South Pole to capture the secretive particles called neutrinos. The result was the IceCube Neutrino Observatory, which can track neutrinos that originate in the distant cosmos.”



FRANCIS HALZEN, 82
University of Wisconsin-Madison, Madison, US.

ON WEDNESDAY:
NOBEL PRIZE IN CHEMISTRY



The IceCube Neutrino Observatory at the Amundsen-Scott South Pole Station, in Antarctica. ICECUBE/NSF

new window for scientists looking at the universe. Extra-galactic neutrinos add another dimension of possibilities.

Together these different kinds of signals enable what is called multi-messenger astronomy, the ability to study the same cosmic event using different kinds of signals, said D Indumathi, a scientist at the Bengaluru-based Indian Institute of Astrophysics who was formerly associated with India's effort to build a neutrino detector.

‘Shy’ particles

The reason neutrinos are so difficult to detect is that they have an extremely low propensity to interact with matter. The Nobel Prize committee described them as the “shiest particle”. They simply pass through very dense regions of matter without any disturbance.

A very common example often cited is the fact that nearly 65 billion neutrinos pass through human finger nails every second without anyone ever noticing. Because they are also electrically neutral, they do

‘Shy’ particles

Neutrinos are difficult to detect because of their extremely low propensity to interact with matter.

• They simply pass through dense regions without disturbance. Nearly 65 billion neutrinos pass through your fingernails every second.

not respond to electromagnetic fields the way charged particles do.

The detections happen on the very rare occasions that they do interact with matter. That happens to, maybe, one in billions or trillions of neutrinos. That is why neutrino observatories the world over are very large facilities, built in extremely isolated, often underground locations, that maximise the chances of interaction and minimise noise.

The elusive nature of neutrinos is also what makes them particularly interesting to study. They pass through regions of cosmic space that other signals may never be able to cross. Very dense regions in space, for example, can absorb or scatter all electromagnetic radiations, but neutrinos simply flow through. They can thus reveal processes happening within or beyond that dense region, which are otherwise inaccessible to scientists.

The IceCube facility

The 82-year-old Halzen, who had been studying cosmic rays since the 1960s, came

to know of the attempts of Soviet researchers to build a neutrino detector in Antarctica, a press release from the Nobel Prize committee said. When a neutrino collides with an atomic nucleus in the ice, it creates an electric spark. This produces radio waves which could be detected using radio antennas.

Halzen thought of a slightly different idea. He realised that light sensors could be installed in glacier ice. These sensors could produce flashes of light from collisions of neutrinos with the nucleus of ice atoms. The area around the South Pole was a favourable location. It was isolated and, at significant depths, there was a lot of darkness with no disturbance from other radiations or signals.

He discussed his ideas with some of his colleagues and did most of the theoretical work on designing an observatory deep below the ice-sheets. IceCube is one cubic kilometer of ice equipped with over 5,000 light sensors on long cables, along with an overground facility where signals are collected. On the rare occasion when the neutrino interacts with an atomic nucleus of ice, a charged particle is produced. If that charged particle travels faster than the speed of light in ice, it produces a faint glow. From the characteristics of this glow, scientists are able to reconstruct the energy and direction of the neutrino involved in the collision.

This exquisite set-up became operational in 2011. By that time, it had already recorded a couple of events that suggested the involvement of extra-galactic neutrinos. In 2013, the observatory presented the first evidence of having detected very high-energy neutrinos. Since then several such incidents have been recorded and verified.

“It’s a very special detector and the work that Halzen has done is really very exciting. It has opened new avenues for astronomy and astrophysics,” said Naba K Mondal, former project director of India-based Neutrino Observatory (INO), a much-delayed effort to build a neutrino detector in India. “In neutrino research circles, this award to Halzen was expected for a few years.” INO was supposed to be located initially in Kerala and then in Tamil Nadu but ran into opposition from local and political groups over land acquisition and environmental grounds. A new location is yet to be finalised.

• ECONOMICS

G7 to release 100 mn barrels of crude, oil products: Will this curb diesel prices?

Sukalp Sharma

New Delhi, October 6

LATE LAST week, the Group of Seven (G7) countries agreed to release 100 million barrels of crude oil and petroleum products from emergency reserves, including substantial volumes of diesel, amid apparent pressure from US President Donald Trump.

With diesel prices at US pumps reaching record levels, Trump had been pressuring Europe to release its diesel stockpiles, and even warned that he could ban diesel exports amid a global supply crunch and soaring prices of the fuel.

Following the G7 announcement, Trump — who wants to lower diesel prices ahead of the midterm polls in the US in November — announced that the country won’t ban exports of its diesel. Interestingly, Trump added that a diesel export ban was never really on the table, although he himself had publicly stated over the past few days that he was inclined towards it.

Why diesel prices are rising

US and European diesel prices surged in recent weeks amid the highly constrained fuel movement through the Strait of Hormuz as well as attacks on Russia’s oil and gas infrastructure.

Diesel is among the most important petroleum fuels globally due to its use in transportation, agriculture, construction, and other industries. This also makes it the most politically sensitive fuel amid the ongoing geopolitical crises.

While G7’s steps won’t do much to deal with the structural problems that the diesel market is grappling with, they are expected to put some downward pressure on international diesel prices, at least in the short-term, according to industry experts.

The extent and duration of relief will largely depend on whether global diesel flows from West Asia and Russia improve hereon.

DIESEL EXPORT BAN WON’T HELP U.S.

• The US is a major diesel exporters. These have surged amid the supply squeeze from West Asia and Russia.

• The fuel’s prices in US are spiking. But restricting diesel exports, as Trump threatened, wouldn’t have helped.

• That’s because of regional disparities within US in production and supply. Also, US prices are linked to international ones.

hit for this segment. Any cool-off in diesel prices will also reduce these companies’ heavy under-recoveries on retail fuel sales. If bulk diesel rates decline, it would be a positive for the Indian economy as inflationary pressure would be eased to some extent.

But higher international supply could squeeze the supernormal margins of standalone Indian refiners — who don’t have much exposure to retail fuel sales — as well as large diesel exporters, which are predominantly private sector refiners like Reliance Industries and Nayara Energy.

The G7 decision: Shoring up supply

The G7 last week agreed to coordinate, through the International Energy Agency (IEA), the release of 100 million barrels of crude oil and petroleum products over four months, including “front-loaded substantial diesel release” in the first 20 days. The countries also agreed to examine additional diesel releases if required by market conditions.

The G7 wants a substantial quantity of diesel to reach the market within days, instead of being spread out over the four-month period, which shows that the tightness in diesel flows is among the most immediate pressure points in the global energy market. But there is a limit to what reserve releases can accomplish on their own. They are not endless and their release can at best be a temporary solution.

Europe has become increasingly dependent on diesel imports from beyond the region over the past few years as it moved away from Russian fuel amid the war in Ukraine. The West Asia war has further exacerbated the supply situation for Europe, which now depends heavily on American diesel.

• HISTORY

China, Russia: The two October revolutions and their enduring legacy



EXPERT EXPLAINS

SYED ALI NADEEM REZAVI

PROFESSOR OF HISTORY AT
ALIGARH MUSLIM UNIVERSITY

OCTOBER HOLDS a singular place in the history of the modern world. It is associated with two revolutions that transformed not merely governments but the very language of politics. On October 1, 1949, Mao Zedong stood at Tiananmen in Beijing and proclaimed the People’s Republic of China. Thirty-two years earlier, in October 1917 (according to the calendar then used in Russia), the Bolsheviks under Vladimir Ilyich Lenin had seized power in Petrograd.

The two revolutions arose from very different circumstances. Yet an important historical thread connects them. The Russian Revolution established the first durable state founded in the name of Marxist social-

ism. The Chinese Revolution demonstrated how Marxism could be reshaped to suit the circumstances of a predominantly agrarian Asian society. Together they profoundly shaped the 20th century and the struggles of millions far beyond their borders.

Russia, 1917

The Russian Revolution cannot be separated from the catastrophe of World War I. Imperial Russia entered the conflict burdened by social inequality, agrarian tensions, a restless industrial working class and an autocracy that had survived the revolution of 1905 without resolving its underlying problems. Military defeats, shortages and inflation eroded the authority of Tsar Nicholas II.

The February Revolution of 1917 brought down the Romanov dynasty, but the Provisional Government continued the war and postponed land reform. Meanwhile, “soviets” (councils) of workers and soldiers emerged as alternative centres of authority. Lenin returned from exile in April and demanded peace and “all power to the soviets”. By September, the Bolsheviks had majorities in the Petrograd and Moscow Soviets, and Leon Trotsky headed the Petrograd Soviet.

In October, Bolshevik-led soldiers,

sailors and Red Guards occupied strategic points in Petrograd and took the Winter Palace. The new government issued decrees on peace and land. To its supporters, October represented the transfer of power to workers and peasants; to its critics, it was the seizure of power by a disciplined minority that eventually led to one-party rule.

Yet, 1917 changed the world far beyond Russia. Land redistribution, workers’ rights, public ownership, mass education and social welfare suddenly appeared as programmes that a state might attempt to realise. Even societies opposed to communism had to respond to the challenge.

China, 1949

China followed another road. The Qing dynasty had fallen in 1911, but political fragmentation, warlordism, unequal treaties and foreign intervention continued. The May Fourth Movement of 1919 radicalised a generation of Chinese intellectuals. The Communist Party of China was founded in 1921, partly under the intellectual impact of the Russian Revolution.

After Chiang Kai-shek turned against the communists in 1927, the movement was driven increasingly into the countryside.



The People’s Liberation Army enters a Chinese city in 1949. WIKIMEDIA COMMONS

Here lay the great Chinese innovation. In a predominantly agrarian society, Mao placed the peasantry at the centre of revolutionary strategy. Rural organisation, land reform and guerrilla warfare became instruments of mobilisation.

The Long March of 1934-35 (the retreat of the communist forces) ensured the survival of the movement. Resistance to Japanese invasion subsequently enhanced the communists’ nationalist credentials, while

renewed civil war after 1945 ended with the defeat of the Kuomintang, the nationalist Chinese party, and its retreat to Taiwan.

On October 1, 1949, Mao proclaimed the People’s Republic. China altered the balance of the emerging Cold War, but its importance to the colonial and post-colonial world went further. It appeared to demonstrate that an agrarian Asian society could develop its own path to revolution. The peasant could become as important a revolutionary figure as the industrial worker.

The meaning of October

The Soviet experiment achieved remarkable advances in literacy, education, science and industrialisation, and the Soviet peoples bore an enormous burden in defeating Nazi Germany. Yet, forced collectivisation, famine, the Great Terror and the Gulag produced immense suffering.

China transformed landholding, education, public health and the legal position of women, and subsequently achieved striking improvements in literacy and life expectancy. Yet, the violence accompanying land reform, the disastrous ‘Great Leap Forward’ and the upheavals of the ‘Cultural Revolution’ also form a part of its history.

Their impact upon colonised societies was nevertheless profound. European imperialism had long presented itself as almost permanent. Russia’s revolution challenged imperialism ideologically, while China showed an Asian civilisation reclaiming political sovereignty.

India responded closely to both experiences. M N Roy debated colonial strategy with Lenin, Bhagat Singh read socialist literature, and Jawaharlal Nehru’s 1927 visit to the Soviet Union influenced his thinking about planning and development.

The Soviet Union disappeared in 1991, while the People’s Republic survives in a form profoundly altered by the economic reforms begun under Deng Xiaoping. Yet the significance of these revolutions extends beyond the states they created. They forced the world to confront questions of land, labour, poverty, inequality, education, welfare, imperialism and national sovereignty.

One need neither romanticise revolution nor overlook its tragedies to recognise its historical power. The political worlds they created have changed enormously, but the questions they raised remain with us.

FULL VERSION ON
WWW.INDIANEXPRESS.COM

Govt announces ANMOL scheme expansion, doubles ASHA incentive

Express News Service
New Delhi, October 6

EXPANSION OF the ANMOL scheme, which raises the free newborn screening programme for newborns, from 56 to 63 diseases, doubling of the monthly incentive for ASHA workers from Rs 3,000 to Rs 6,000 — these were among the several new healthcare initiatives launched by the Delhi Government on Tuesday under the “Seva Sankalp Abhiyan” programme, organised at Thyagraj Stadium.

The programme also saw the virtual inauguration of the State Organ and Tissue Transplant Organisation (SOTTO) building at GB Pant Hospital.

Further, a Nursing Home Management System portal was launched, which moves registration and renewal for private nursing homes online, officials said.

The hike in incentives is set to benefit 6,690 ASHA workers in the Capital, officials said, adding that the increment is a recognition of the dedication with which the workers bring Delhi’s health services to households. ASHA workers act as the link between the community and the public health system. They register pregnant women, push for institutional deliveries, track immunisation, counsel families on nutrition and sanitation, and accompany patients to health facilities and



Chief Minister Rekha Gupta during the launch of the healthcare initiatives in New Delhi on Tuesday. PTI

work under the National Health Mission.

Speaking on the develop-

ment, Health Minister Pankaj Kumar Singh said, “The incentive had been pending for

nearly a decade. Through the Covid years, the workers kept serving. The steps to recognise them did not come. This government has addressed it.”

On Tuesday, Chief Minister Rekha Gupta also felicitated outstanding workers at the “ASHA Sammelan-2026 and ASHA Samman Samaroh”. She said she expects them to reach every family like an elder sister, and announced a special dandiya programme for them during Navratri.

Delhi Lieutenant Governor Taranjit Singh Sandhu said, “ASHA workers stand on the front lines of the healthcare system, taking health services and awareness to pregnant women, children and families.”

On the expansion of Mission ANMOL, officials said that it now covers 63 congenital health conditions, with screening to be carried out within 24 to 72 hours of birth.

Since Mission ANMOL was made operational in March this year, more than 75,000 newborns have been screened and over 700 children have been treated in time, officials said.

The government aims to increase the annual coverage under the scheme around 1.5 lakh to 2.5 lakh newborns. Last month, The Indian Express had reported that more than 14,000 newborn genetic samples arrive every month at the LNJP genetic lab, the only genetic lab run by the Delhi government.

EXPRESSspecial

PCI operating at half strength, without a single working journalist or newspaper editor

I&B Ministry says notification for 13 more members to be issued soon

Amrita Nayak Dutta
New Delhi, October 6



Justice Ranjana Desai was reappointed PCI chairperson in April this year

THE COUNTRY’S key media watchdog, Press Council of India (PCI), has been operating at half its sanctioned strength for nearly a year now and without a single working journalist or newspaper editor.

The Ministry of Information and Broadcasting (I&B) has not yet formally notified the names of 13 of the 28 PCI members, six of whom should be newspaper editors and seven should be working journalists other than editors, despite the PCI nominating them earlier this year.

Ironically, the PCI remained without a chairman for nearly four months until Justice Ranjana Prakash Desai, a retired Supreme Court judge, was reappointed to the post in April this year. Justice Desai had completed her previous tenure in December last year, resulting in a prolonged vacancy before her reappointment.

Responding to queries from *The Indian Express* on the matter, the I&B Ministry said it is informed that the notification of the names of the 13 members is currently at the final stage and is expected to be issued shortly, upon completion of the requisite formalities.

The PCI operates under the Press Council Act, 1978, which stipulates that the Council must consist of a chairperson and 28 members, each serving a three-year term.

The members include six newspaper owners, one representative from news agencies, five Members of Parliament and

three persons with special knowledge or practical experience in education, science, law, literature and culture, nominated by the University Grants Commission, the Bar Council of India and the Sahitya Akademi, respectively. In addition, the Council includes six newspaper editors and seven working journalists. The Act therefore mandates the appointment of 13 newspaper editors and working journalists.

However, after the 14th Press Council completed its term in October 2024, the 15th Press Council has yet to be fully constituted. At present, only 14 of the 28 members have been appointed: five MPs, three nominees and six from among newspaper owners and management, aside from the chairperson.

Documents accessed by *The Indian Express* show that the PCI has nominated the 13 members from its side in batches since last year, under Section 5(3)(a) of the Press Council Act. In December last year, the PCI nominated six newspaper editors, followed by six working journalists in May this year. The seventh working journalist was nominated in July.

The documents further show that the PCI sent three communications to the I&B Ministry separately — dated May 25, July 2 and September 9 — requesting it to issue the Ga-

zette notification under Section 5(5) of the Act.

The section states that the Central Government shall notify the names of persons nominated as members in the Official Gazette, and that each nomination shall take effect from the date of notification.

In its latest communication to the Ministry, dated September 9, the PCI requested that the Gazette notification for the 13 members be issued at the earliest. It said it had received communications from nominees expressing concern over the delay in officially notifying their appointments as PCI members.

In the earlier communication dated July 2, the PCI asked the Ministry to expedite the notification of the 12 individuals it had nominated to the Council by that date.

“This is essential to enable the 15th term of the Council to become fully representative and functional so that it can discharge its regular responsibilities,” the letter stated. The matter, it is learnt, is under process at the Ministry’s end.

The PCI has continued to hold meetings despite functioning with half its sanctioned strength. It held its last meeting in June, and another is scheduled for later this month.

The issue was also raised in Parliament in February by BJD legislator Sasmit Patra, who highlighted the serious implications of the absence of a duly constituted Council and a chairperson. He urged the government to treat the matter with the seriousness it deserved and take immediate steps to appoint a chairperson and constitute the 15th Press Council.

“Restoring the full functioning of the PCI is essential to uphold democratic values, strengthen media accountability and reaffirm the government’s commitment to a free, fair and responsible press,” he said.



~ FOUNDED BY ~

RAMNATH GOENKA

◆ IN 1932 ◆

BECAUSE THE TRUTH INVOLVES US ALL

Three years later, a peace still out of reach

THE HORRIFIC terror attack on Israel by Hamas three years ago, in which almost 1,200 were killed and 251 taken hostage, set in motion a war that continues to ripple far beyond Gaza. The devastation that followed has left Israel isolated internationally, with several Western nations formally recognising the State of Palestine. Prime Minister Benjamin Netanyahu's decision to conduct joint military operations with the US against Iran, perceived to be weakened by internal turmoil and the decimation of its proxies, has plunged West Asia into further chaos. No lasting settlement in Gaza is possible without Israel's participation, or without Hamas giving up arms — peace still doesn't look within reach.

The situation in Gaza, where more than 70,000 people have been killed in the last three years, continues to be enormously disturbing. The humanitarian crisis remains acute, with shortages of supplies, displacement and military strikes by the Israel Defence Forces (IDF) despite the ceasefire that began almost a year ago. The much-touted peace plan of US President Donald Trump has made little progress. Only two weeks ago, the Board of Peace announced a \$2.45 billion recovery plan for the Strip — a fraction of the \$71.4 billion that the UN estimates will be required for recovery and reconstruction over the next decade. Hamas's disarmament remains an issue locked in an impasse: Israel says reconstruction cannot proceed without disarmament, while Hamas, which prolongs the stalemate and the suffering of Gazans, resists disarmament without a full Israeli withdrawal first. The first phase of the Trump plan envisaged Israel retaining control of 53 per cent of Gaza; since then, however, IDF forces have expanded their presence to two-thirds of the Strip.

The power to enforce meaningful progress lies ultimately with the White House, but the US President seems consumed by the political fallout of the war with Iran. At the same time, the once-unbreakable US-Israel alliance is fraying: Public support for Israel in America has fallen to historic lows, and the bipartisan consensus around military aid shows signs of weakening. Within Israel, the Gaza war has exposed deep divisions over the government's conduct — especially in light of revelations that Netanyahu ignored warnings from Egypt and the UAE about an impending Hamas attack — and the country's entanglement in regional conflicts, including in Lebanon and Syria. In 20 days, Israel will hold an election, giving citizens their first chance since the terror attack to pronounce a verdict on Netanyahu. The coalition that emerges after October 27 will help shape not only the future of Gaza, but also Israel's trajectory. It will indicate whether three years of militarism can give way to a path that leads away from war.



A war in Iran, an election in America, and the blurred lines between politics & conflict



RAJA MANDALA
BY C RAJA MOHAN

IRAN'S REVOLUTIONARY Guards have found a new audience for their war with Donald Trump: The American voter. The history of warfare tells us that weakening the adversary's domestic support can be as important as defeating its armed forces on the battlefield. In a 25-page letter last week, the Guards appealed to the "honourable people" of the United States to reject Trump's policies and consider "peaceful coexistence" with Iran. Trump, in turn, hopes that escalating military and economic pressure will force the Islamic Republic to fall in line.

Coming just weeks before the midterm elections, the letter brings Tehran directly into America's domestic political contest. With Republicans vulnerable in November, the political consequences of the Iran war are becoming harder to separate from its military course. In Tehran, too, internal struggles between advocates of negotiation and hardliners favouring confrontation are shaping the war's conduct. That brings us to the idea of the "October surprise" in American politics — a major event or political move that changes the national mood before the traditional voting in the first week of November.

Iran helped make the phrase part of American political language during the 1980 presidential election. The Iranian Revolution of February 1979 overthrew the Shah, a close American ally. In November, militants seized the US embassy in Tehran and took its staff hostage. The crisis haunted President Jimmy Carter's re-election campaign. His Republican challenger, Ronald Reagan, feared that freeing the hostages before November would rescue Carter's political fortunes. Allegations that Reagan's team secretly worked to delay the release remain disputed. The hostages came home as Reagan took office in January 1981. Iran's up-

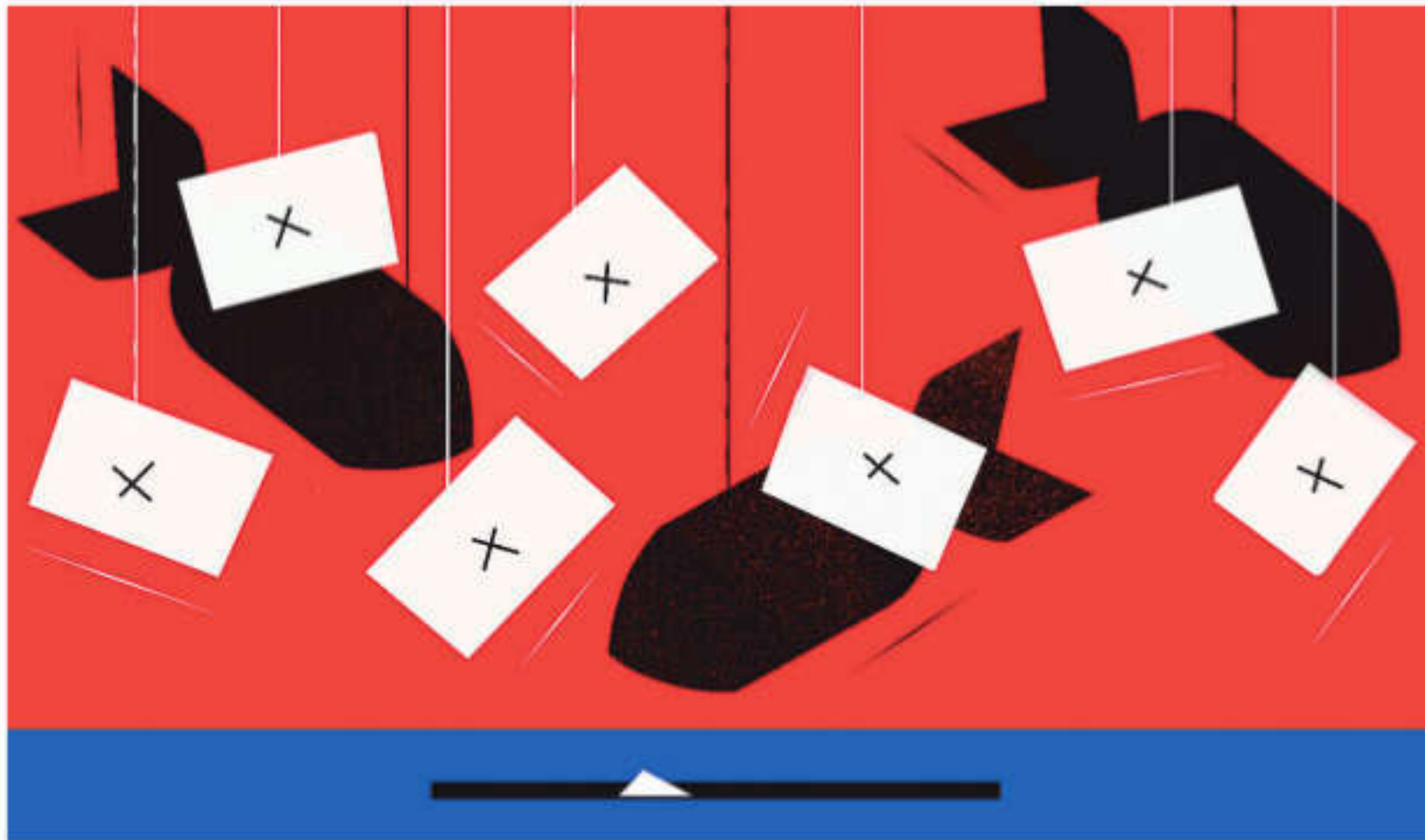


ILLUSTRATION: C R SASIKUMAR

heaval had become entangled with the struggle for power in Washington.

Today, Tehran hopes to exploit another American electoral moment. Trump insists the midterms do not determine his Iran policy. But the electoral calendar is now visibly intertwined with the military debate. In an interview last week, Trump said it was "possible" that US bombing of Iran could intensify after the November 3 elections. He has also said the war could be won "right after the election" — or perhaps before it. The military preparations reinforce the message. The unfolding deployment gives Trump more options if diplomacy fails — and gives Tehran reason to take his threats seriously.

Setting aside the Islamic Revolution's enduring slogan, "Death to America", the IRGC letter claims to have no quarrel with ordinary Americans. Its immediate target is Trump. It hopes public anxiety over the war and rising prices will weaken Republicans and create space for a more accommodating policy in Washington. The appeal could easily backfire as Republicans accuse Democrats of serving Tehran's interests.

Iran is not alone in trying to exploit domestic divisions. Trump

has hoped that military and economic pressure would encourage Iranians to challenge their rulers. That expectation has produced little so far. Tehran's appeal to American voters may prove equally ineffective. Both sides nevertheless see the adversary's home front as a battlefield.

Meanwhile, Trump has secured an important gain at Hormuz. The recovery in Gulf oil flows has accelerated. New data show Gulf crude exports averaged about 16.3 million barrels a day in September — roughly 91 per cent of pre-war levels.

This does not establish Trump's claim of "total control". Iran never enjoyed uncontested sovereignty over the international waterway. Its leverage lay in making passage dangerous and expensive. Tanker attacks, higher insurance costs and uncertainty forced its adversaries to spend heavily on keeping commerce moving. Nor has the energy crisis disappeared. The problem has shifted from crude supply towards logistics and refining. Refined-product exports from the Gulf recovered to only about 60 per cent of pre-war levels in September.

That distinction matters politically. More crude moving through

Hormuz does not automatically mean cheaper diesel at American pumps. Crude must be transported, refined and distributed. Diesel prices have a direct effect on inflation and cost of living, a major issue in US domestic politics right now.

The G7's decision, under Trump's pressure, to release 100 million barrels of crude and fuel products from emergency reserves is meant to ease this pressure. But Trump's dilemma remains. Convincing Tehran that it cannot wait him out requires a credible threat of escalation. Reassuring American voters requires avoiding an escalation that sends fuel prices soaring again. Iran's letter seeks to exploit precisely that tension.

Tehran faces pressures of its own. Washington's "Operation Economic Outcast" is tightening the financial noose. The US has expanded sanctions from military procurement networks to Iran's automotive, rail, manufacturing and steel sectors. Washington has now warned foreign financial institutions that doing business with Iran could bring sanctions without prior notice.

Neither side can assume the other will crack first. The October surprise could be another military blow, a diplomatic opening, or another fuel-price shock.

For India, there is a larger lesson from the US and Iranian efforts to influence each other's domestic politics. America's hopes of political change in Iran and Tehran's appeal to American voters underline an old truth: Wars are fought within societies as much as between states. Managing the home front can matter as much as gaining military ground on the battlefield.

The principle applies in peacetime too, especially to democracies such as India. Effective engagement with a turbulent world demands cohesion at home. Political differences are inevitable in a democracy. But a politics of "winner takes all" can turn those differences into debilitating domestic conflicts. That inevitably weakens the nation's capacity to deal with mounting challenges abroad and the growing prospects for external meddling in its domestic affairs.

The writer is contributing editor on international affairs for The Indian Express. He is professor and director at the Motwani Jadeja Institute of American Studies, Jindal Global University, Delhi



**SRIRAM BALASUBRAMANIAN
AND PRAKASH LOUNGANI**

THE DEVELOPMENTS dominating global markets are the 10-year US treasury yields rising above 5 per cent and the US Federal Reserve's rate hike cycle to contain inflation. These matter for India, shaping challenges and policy choices.

First, the 10-year treasury yield has risen to its highest level since 2007. The immediate impact is a narrowing of the India-US yield spread to around 200 basis points, limiting the scope for further inflows compared with periods when spreads were wider. Higher yields also raise fiscal risks by increasing borrowing costs, especially for

long-tenor issuances that lock in elevated costs for decades. Over the longer term, they point both to concerns about US fiscal and debt sustainability and to uncertainty around the AI infrastructure boom. They may also signal a broader "higher for longer" environment, closer to the pre-2000s market setting. India will therefore need to navigate these cross-currents carefully.

On the issue of the Fed's rate hikes, the question now is: How long will this cycle last? Since inflation in the US is closer to the 2 per cent target than in the previous cycle, this one may be shorter. Even so, short-end rates are likely to stay elevated, reinforcing yield pressures through spreads, currency, and liquidity. But, India's economy is much more resilient than in earlier episodes of high yields and rising rates in the US. The 10-year sensitivity has fallen sharply from 1.25 in 2013

to 0.43 in 2026. This points to an economy more resilient to external volatility. What explains this, and why does it matter?

Several factors explain this resilience. India's macro fundamentals are perhaps at their strongest in decades: The fiscal deficit is on a glide path toward 4 per cent, inflation is within the target band, the current account is stable, and reserves remain ample especially after the FCNR (B) scheme. Retail participation has risen sharply, while domestic institutional investors have overtaken foreign institutional investors in

Considering the high yields, the possibility of an AI bubble bursting is a distinct possibility, and therefore an important risk

ownership, deepening India's domestic capital base and therefore reducing the impact on capital flow volatility. There is palpable dynamism in the economy that has been the largest contributor to global growth in the last few years. These strengths provide comfort, though important concerns remain.

First, considering the high yields, the possibility of an AI bubble bursting is a distinct possibility, and therefore an important risk. India's exposure should be analysed carefully. Second, the US-Iran geopolitical conflict remains a major constraint for India, especially with respect to the price of oil and a variety of other goods imported from the Middle East. Each spike in Brent crude oil prices raises imported inflation, weakens the Indian rupee, and pushes yields higher. With the US Fed hiking, India has little room to absorb that pressure. Re-

lief can also come quickly, as when Brent fell to about \$98 per barrel and the 10-year bond yield eased toward 7 per cent. But even then, Fed tightening could keep a floor under yields. Third, state financing through the bond market needs to be monitored carefully, especially given that many of them tend to borrow in the later part of the year. State borrowing costs have risen with recent SDL auctions clearing between roughly 7.3 per cent and 7.9 per cent.

These risks need to be carefully monitored. But with strong macroeconomic fundamentals, India does have the capacity to withstand changes in the global environment, if it plays its cards well.

Balasubramanian is an economist and Loungani is director of the Applied Economics Programme at Johns Hopkins. Views are personal

Sujit Bisoyi
Bhubaneswar, October 6

He also highlighted proposed investment of around Rs 5,000 crore, indigenisation of raw materials in partnership with Mishra Dhatu Nigam Limited (MIDHANI), and establishment of India's first indigenous high-thrust engine



testing facility.

Emphasising the need for reliable indigenous propulsion for India's helicopter ecosystem, the defence minister stated that forces require engines that are available when needed, supported throughout their lifecycle, and upgraded with evolving technology.

The defence minister announced that HAL is enhancing annual AL-31FP engine production capacity from 30 to 50 engines annually with an investment of Rs 335 crore. He also announced a second Shakti engine production line with an investment of Rs 218 crore.

UNION CABINET ANNOUNCES A HOST OF MEASURES TO BOOST GROWTH

Small, medium enterprises get Rs 10,000 cr fund boost

Pratyush Deep &
Ravi Dutta Mishra
New Delhi, October 6

THE UNION Cabinet on Tuesday approved the Small and Medium Enterprise (SME) Growth Fund (SGF) with an outlay of Rs 10,000 crore.

This fund to provide equity support to small and medium enterprises was first announced in the Union Budget for 2026-27.

According to the government, while several existing funds provide equity support to businesses, most of them focus on early-stage enterprises and largely cater to micro enterprises. This has left a structural gap in the availability of growth-stage equity capital for small and medium enterprises.

"While various initiatives have enhanced access to credit for SMEs, a gap remains in the availability of long-term risk capital required by enterprises seeking to scale, innovate, expand internationally, adopt advanced technologies, undertake acquisitions, and transform into industry leaders," the government said in a statement.

"The SME Growth Fund is designed to address this critical financing gap by providing patient growth equity capital to high-potential SMEs with demonstrated business viability and scalability," it added.

Under the initiative, the government will provide an aggregate commitment of Rs 10,000 crore to the Alternative Investment Fund (AIF) established under the SGF framework. A majority of the fund's investments will be directed towards small and medium manufacturing enterprises, with a focus also on SMEs operating in industrial clusters in Tier-II and Tier-III



A majority of the fund's investments will be directed towards small and medium manufacturing enterprises, with a focus also on SMEs operating in industrial clusters in Tier-II and Tier-III cities. REUTERS

cities. "Investments across industrial clusters, including those in Tier-II and Tier-III cities, will support balanced regional industrial development, reinforce local supply chains and generate high-quality employment opportunities," it said.

The Centre expects the fund to help SMEs scale up, improve productivity and strengthen export competitiveness by enabling manufacturing enterprises to expand capacity, adopt advanced technologies and achieve greater scale.

India SME Forum in a note said the fund is critical for the MSME as they are facing structural and severe capital gaps. They have been over-dependent on debt financing and have weak integration with the global value chain, the forum said.

"The SME growth fund could help mobilisation of institutional capital and creation of globally competitive mid-sized champions," it said.

Vinod Kumar, President, India SME Forum said the Rs 10,000 crore SME Growth Fund can set in motion a powerful cycle, where stronger manu-

facturers place larger and more consistent orders with smaller suppliers, enabling those businesses to invest in better machinery, improve quality, build skills and create jobs.

"As these capabilities deepen across industrial clusters, more MSMEs can meet the standards required by large domestic buyers and global supply chains. The cumulative outcome can be higher domestic value addition, greater export competitiveness and reduced dependence on imported components and finished products," Kumar said.

Experts said the fund is critical for the MSME as they are facing structural gaps and severe capital gaps and have been over-dependent on debt financing and have weak integration with the global value chain. Sandeep Jain, an auto-component manufacturer and former President of Federation of Indian Micro and Small & Medium Enterprises (FISME) said the growth fund would give motivation to SMEs to grow and not be overtly concerned with the financing.

FULL REPORT ON
WWW.INDIANEXPRESS.COM

New apex body for transport and logistics gets nod

New Delhi: The Union Cabinet on Tuesday approved the constitution of an Integrated Transport & Logistics Authority (ITLA) to have a synergy among different transport departments and assess the country's future requirements in infrastructure development.

The authority will be set up in the Ministry of Commerce & Industry, serving as the apex body for project planning and appraisal, research, monitoring and evaluation, among other functions.

Minister of Information and Broadcasting Ashwini Vaishnaw said that all the major sectors such as the railways, highways, aviation, among others, will have representation under the new authority.

"Practically, if GDP increases by 7%, the logistics requirement increases by 10% in the economy. To meet this heavy requirement, Highways, railway lines, waterways, ports, airports are being built. To have a unified planning of these things, Gati Shakti portal was launched. Now, to meet the future requirements in transportation and have a synergy among different departments, the Integrated transport & logistics authority will be set up," said Vaishnaw, as he announced the decision. ENS

FULL REPORT ON
WWW.INDIANEXPRESS.COM

World Bank raises India's FY27 growth outlook to 7.1%

Siddharth Upasani
New Delhi, October 6

THE WORLD Bank has raised its GDP growth forecast for India for the current fiscal by half a percentage point to 7.1%, with this rapid expansion seen driving the broader South Asia region. According to the multilateral organisation's twice-yearly South Asia Economic Update, released on Tuesday, the South Asia region's GDP in 2026 is seen expanding by 6.9%, up from the previous forecast of 6.3%.

However, excluding India, regional growth is seen 3.6%, down from 4.1% earlier, mostly because of a large downward revision for Bangladesh. The World Bank expects the "strong" momentum in industry and services to continue, with the reforms that have been undertaken — consolidation of labour codes, GST reforms, rationalisation of tariffs, the bankruptcy and insolvency act, and physical and digital infrastructure investments — keeping the growth potential strong. "Strength in the rest of the economy is expected to make up for agricultural weakness," the World Bank said, adding that the sub-par monsoons will likely reduce agricultural output.

FULL REPORT ON
WWW.INDIANEXPRESS.COM